Taking vitamin D with the largest meal improves absorption and results in higher serum levels of 25-hydroxyvitamin D.

Many patients treated for vitamin D deficiency fail to achieve an adequate serum level of 25-hydroxyvitamin D [25(OH)D] despite high doses of ergo- or cholecalciferol. The objective of this study was to determine whether administration of vitamin D supplement with the largest meal of the day would improve absorption and increase serum levels of 25(OH)D. This was a prospective cohort study in an ambulatory tertiary-care referral center. Patients seen at the Cleveland Clinic Foundation Bone Clinic for the treatment of vitamin D deficiency who were not responding to treatment make up the study group. Subjects were instructed to take their usual vitamin D supplement with the largest meal of the day. The main outcome measure was the serum 259(OH)D level after 2 to 3 months. Seventeen patients were analyzed. The mean age (+/-SD) and sex (F/M) ratio were 64.5 +/- 11.0 years and 13 females and 4 males, respectively. The dose of 25(OH)D ranged from 1000 to 50,000 IU daily. The mean baseline serum 25(OH)D level (+/-SD) was 30.5 +/- 4.7 ng/mL (range 21.6 to 38.8 ng/mL). The mean serum 25(OH)D level after diet modification (+/-SD) was 47.2 +/- 10.9 ng/mL (range 34.7 to 74.0 ng/mL, p < .01). Overall, the average serum 25(OH)D level increased by 56.7% +/- 36.7%. A subgroup analysis based on the weekly dose of vitamin D was performed, and a similar trend was observed.Thus it is concluded that taking vitamin D with the largest meal improves absorption and results in about a 50% increase in serum levels of 25(OH)D levels achieved. Similar increases were observed in a wide range of vitamin D doses taken for a variety of medical conditions. Copyright 2010 American Society for Bone and Mineral Research.